Architecture Diagrams — Detailed Briefing

All 4 Customer References — AWS CloudOps Competency Audit

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INTRODUCTION

This document provides a detailed briefing of the architecture diagrams for all four customer references submitted as part of the AWS CloudOps Competency audit. Each architecture is described in full — covering the AWS services used, how they are deployed (VPCs, AZs, subnets), how the design scales automatically, how it achieves high availability, and how major components recover from failure.

The four customer references are:

- Customer Reference #1: Franciscan Solutions (EdTech) — School ERP & Mobile Platform
- Customer Reference #2: Rev (Web3/Digital Services) — Blockchain Validator & dApp Platform
- Customer Reference #3: Genepowerx (HealthTech) — AI Medical Imaging & Diagnostics
- Customer Reference #4: KiwiKisan (AgriTech) — IoT Sensor Data Pipeline & Analytics

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ARCHITECTURE 1: FRANCISCAN SOLUTIONS (EdTech)

Overview

Customer: Franciscan Solutions

Industry: Education Technology (K-12 School ERP & Mobile Platforms)

Engagement: FinOps Transformation — Cloud Financial Management

AWS Account Structure: Multi-account with Production, Non-Production, and Shared Services OUs

Franciscan runs a digitized ERP and mobile platform serving K-12 schools. The architecture is designed to handle predictable school-year traffic patterns with seasonal spikes during admissions and exams, while maintaining cost efficiency during off-peak periods.

AWS Services Used

AWS Service	Role in Architecture
Application Load Balancer (ALB)	Entry point for all user traffic. Deployed across multiple Availability Zones for high availability. Automatically scales capacity in response to incoming application traffic.
Amazon EKS (Elastic Kubernetes Service)	Hosts the core ERP and mobile application workloads as containerized microservices. The EKS control plane is managed by AWS across multiple AZs. Worker nodes run in Auto Scaling Groups distributed across multiple AZs.
Amazon RDS (Aurora Serverless)	Database layer for the Student Portal and ERP data. Configured as Aurora Serverless with capacity scaling up to 12.0 ACUs. Auto-pause enabled to scale down during school holidays, reducing unit cost per student. Multi-AZ deployment with automatic failover.
Amazon S3	Object storage for documents, reports, and static assets. Lifecycle policies transition infrequently accessed data to cheaper storage classes. Inherently regional and multi-AZ by default.
AWS Lambda	Serverless compute for event-driven workloads (e.g., report generation, notifications, automated remediation). Scales from zero to thousands of concurrent executions automatically.
Amazon CloudFront	CDN for content delivery across regions. Caches static assets at edge locations for low-latency access by schools across geographies.
Amazon DynamoDB	NoSQL database for high-throughput, low-latency application data (e.g., session management, real-time activity tracking).
Amazon Athena	Serverless query engine for analyzing CUR data stored in S3. Used for cost analysis and unit economics calculations.
AWS Cost Explorer & AWS Budgets	Cost trend analysis, predictive estimates, budget alerts with email and webhook notifications, anomaly detection.
AWS Organizations	Centralized management, consolidated billing, Tag Policies, and SCPs for governance across all accounts.
AWS Config	Continuous compliance scanning with managed and custom rules for tag enforcement.
Amazon QuickSight	Cloud Intelligence Dashboards (CID) for cost-per-DAU tracking, service-level breakdowns, and executive reporting.
Aquila Cloud FinOps Platform	AI-driven cost analysis, anomaly detection, optimization recommendations, tag compliance tracking, right-sizing, RI/SP management.

High Availability & Resilience

[How to explain this to the auditor]

"This architecture prioritizes high availability and cost-effective scaling to handle predictable school-year traffic patterns."

Application Load Balancer (ALB)
Deployed across multiple Availability Zones. If one AZ fails, the ALB nodes in the healthy AZs automatically handle all incoming traffic. No manual intervention required.

Amazon EKS Cluster
The EKS control plane is managed by AWS across multiple AZs — this is inherent to the managed service. Worker nodes are deployed in an Auto Scaling Group distributed across multiple AZs. If an AZ goes down, Kubernetes automatically reschedules application pods onto nodes in the healthy AZs. There is no single point of failure at the compute layer.

Amazon RDS (Aurora Serverless)
Configured in a Multi-AZ deployment. In case of a primary database failure, AWS automatically fails over to the synchronous standby replica in a different AZ with no data loss. Aurora Serverless additionally provides automatic capacity scaling — scaling up during admission/exam peaks and pausing during holidays.

Core Services (S3, Lambda)
These services are inherently regional and multi-AZ by default, providing high availability without extra configuration. S3 replicates objects across multiple AZs within the region. Lambda runs across multiple AZs automatically.

Auto-Scaling & Elasticity

Application Load Balancer
Scales its capacity automatically in response to incoming application traffic. No pre-provisioning required.

Amazon EKS — Dual-Layer Scaling
The EKS cluster uses a dual-layer auto-scaling approach:

- Pod Level (HPA — Horizontal Pod Autoscaler): Automatically increases or decreases the number of application containers based on CPU and memory load. This provides rapid response to traffic spikes — new pods spin up in seconds.

- Node Level (Cluster Autoscaler): Automatically adds or removes EC2 worker nodes from the cluster to ensure there is enough capacity for the pods to run. When HPA creates more pods than existing nodes can handle, the Cluster Autoscaler provisions new nodes.

AWS Lambda

Scales from zero to thousands of concurrent executions automatically based on the number of invocations. No capacity planning required.

Aurora Serverless

Automatically scales database capacity up to 12.0 ACUs based on load. Auto-pause feature scales to zero during idle periods (school holidays), directly reducing cost-per-student.

IaC Implementation

All infrastructure is provisioned via Terraform. The IaC code for Franciscan includes:

- Aurora Serverless with capacity scaling up to 12.0 ACUs and auto_pause = true
- 4 mandatory FinOps tags enforced on all core services (application_id, cost_center, client_id, environment)
- Tag compliance improved to 96% through IaC-enforced tagging

The Terraform code is version-controlled in Git and deployed through a CI/CD pipeline with static analysis (tflint), security scanning (tfsec, Checkov), and tag compliance checks.

Multi-Account Structure

Franciscan's AWS environment is organized using AWS Organizations:

- Production OU — Live ERP and mobile application workloads
- Non-Production OU — Development, QA, and staging environments
- Shared Services OU — Centralized networking, logging, security, and FinOps governance
- Central governance enforced through SCPs, Tag Policies, AWS Budgets, and anomaly detection at the OU level
- A dedicated FinOps account manages CUR, cost governance, and standardized controls

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Disaster Recovery

RTO Target: 4 hours

RPO Target: 1 hour

Recovery leverages Amazon EKS snapshots, AWS Backup, and Multi-AZ deployment. Kubernetes StatefulSets provide data persistence with automated failover. Regular testing of backup and recovery processes is part of the operational review cadence.

ARCHITECTURE 2: REV (Web3 / Digital Services)

Overview

Customer: Rev

Industry: Blockchain & Digital Assets (Web3 Infrastructure)

Engagement: FinOps Transformation — Unit Economics & Cost Governance

AWS Account Structure: Multi-account with Production, Non-Production, and Shared Services OUs (workloads isolated by blockchain/application)

Rev operates a high-performance Web3 infrastructure platform running 24/7 blockchain validator nodes and decentralized application (dApp) backends. The architecture prioritizes global low-latency performance, massive scalability for volatile transaction volumes, and resilience against regional events.

AWS Services Used

AWS Service	Role in Architecture
AWS Global Accelerator	Provides global traffic routing and resilience against regional failures. Automatically reroutes traffic to the nearest healthy regional endpoint if an entire AWS Region becomes unhealthy.
Application Load Balancer (ALB)	Scales automatically to handle fluctuating API traffic from dApps.
Amazon EKS	Hosts dApp backend microservices with dual-layer auto-scaling (HPA + Cluster Autoscaler).
EC2 Validator Nodes	Dedicated compute for blockchain validation. Deployed in Auto Scaling Groups across multiple AZs. Uses Graviton instances (c6g.2xlarge) for 34% cost reduction. Max scaling to 25 nodes.
Amazon DynamoDB	On-demand mode for unpredictable blockchain workloads. Multi-AZ by default with synchronous replication across 3 AZs. Global Tables for active-active multi-region database.
Amazon Managed Blockchain	Managed service with built-in HA. Automatically manages node health and replacement.
Amazon S3	Object storage for blockchain data, logs, and artifacts.
Amazon CloudFront	CDN for dApp frontend delivery.
AWS Lambda	Event-driven serverless compute for automation and remediation.
Amazon Athena	Serverless CUR data querying for cost analysis.
AWS Cost Explorer & Budgets	Cost trend analysis, anomaly detection, budget alerts.
AWS Organizations	Centralized governance with 5 mandatory tags enforced via SCPs.
Aquila Cloud FinOps Platform	AI-driven cost analysis, unit economics (cost-per-transaction), anomaly detection.

High Availability & Resilience

AWS Global Accelerator

Provides resilience against regional failures. If an entire AWS Region becomes unhealthy, Global Accelerator automatically reroutes traffic to the next nearest healthy regional endpoint in seconds. This is critical for 24/7 blockchain operations where downtime means missed transactions.

Compute Tier (EC2 Nodes & EKS)

All compute resources are deployed in Auto Scaling Groups across multiple AZs. An AZ failure does not impact overall availability of validator nodes or dApp backends. Kubernetes reschedules pods automatically.

Amazon DynamoDB

Native multi-AZ service — synchronously replicates data across 3 AZs. For global resilience, DynamoDB Global Tables provide an active-active multi-region database, ensuring data availability even during a full regional outage.

Amazon Managed Blockchain

As a managed service, it includes built-in high availability and automatically manages node health and replacement. No manual intervention required for node failures.

Auto-Scaling & Elasticity

- ALB: Scales automatically to handle fluctuating API traffic from dApps
- EC2 Validator Nodes: Auto Scaling Group with dynamic scaling policies based on network load and blockchain-specific metrics. Max capacity of 25 nodes.
- Amazon EKS (Dual-Layer): HPA for pod-level scaling + Cluster Autoscaler for node-level scaling
- Amazon DynamoDB: On-demand mode — instantly scales read/write capacity without throttling, ideal for unpredictable blockchain workloads

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IaC Implementation

Terraform defines the entire solution:

- max_size = 25 in the scaling config allows dynamic scaling of validator nodes
- Graviton instance types (c6g.2xlarge) mandated in IaC — 34% cost reduction
- 5 mandatory FinOps tags enforced (Owner, Chain, Environment, Application, CostCenter) — 98% tag compliance

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Multi-Account Structure

- Root: Management Account
- OU: Infrastructure — Shared Services (Networking, Logging)
- OU: Production — Production workload accounts (one or more per blockchain/region)
- OU: Non-Production — Dev/Test accounts
- Centralized tagging, budgets, and guardrails enforced through Organizations

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Key Metrics

- 34% reduction in blended monthly AWS costs
- 40% reduction in unit cost (cost per 1M processed transactions)
- Savings Plans coverage optimized from 25% to 85% utilization
- Tag compliance from 45% to 96-98%

ARCHITECTURE 3: GENEPOWERX (HealthTech)

Overview

Customer: Genepowerx

Industry: Healthcare Technology (AI Medical Imaging & Diagnostics)

Engagement: FinOps & HIPAA Compliance Transformation

AWS Account Structure: Multi-account with dedicated OUs for Production, ML/R&D, and Compliance

Genepowerx operates an AI-powered medical imaging and diagnostics platform serving multiple hospital clients. The architecture must maintain strict HIPAA compliance (100% encryption at rest and in transit) while scaling GPU-intensive inference workloads cost-effectively. The design separates PHI-sensitive data, ML training, and inference systems into isolated accounts.

AWS Services Used

AWS Service	Role in Architecture
Amazon SageMaker Endpoints	AI inference endpoint for medical image analysis. Deployed with instances in multiple AZs. Auto Scaling policy monitors invocation rate — adds GPU instances during peak hours, removes during quiet periods.
Amazon SageMaker (Spot Training)	ML model training using Spot instances — 35% cost reduction on training jobs. Separate from inference workloads.
EC2 G5 (GPU Instances)	Specialized GPU compute for AI inference. Auto Scaling Group configured for max 50 GPU instances. Scales rapidly during diagnostic peak usage.
AWS HealthImaging	Specialized service for medical imaging data storage and retrieval. Optimized for DICOM and other medical imaging formats.
Amazon S3 (Intelligent-Tiering)	Storage for medical imaging data with Intelligent-Tiering — automatically moves data to colder tiers. 60% reduction in storage costs. 100% KMS encryption at rest enforced.
AWS Lambda	Orchestration layer — scales based on number of images uploaded. Provisions compute for each invocation automatically.
AWS KMS	Key management for all encryption. Inherently highly available across multiple AZs.
VPC Endpoints	Redundant VPC Endpoints in each private subnet ensure connectivity to AWS services remains available even if one endpoint fails. Keeps all traffic within the AWS network — no internet exposure for PHI data.
Amazon Athena	CUR data querying for detailed financial auditing and compliance verification.
AWS Organizations	OU-level SCPs, encryption standards, tag policies, and automated audit logging.
Aquila Cloud FinOps Platform	GPU utilization analysis, Savings Plans optimization (raised to 80% coverage), unit cost calculation (cost-per-scan).

High Availability & Resilience

Amazon SageMaker Endpoints

Inference endpoint deployed with instances in multiple AZs. If an instance or entire AZ becomes unavailable, SageMaker automatically routes inference requests to healthy instances in the other AZ. Continuous service for hospital clients — no diagnostic downtime.

Core Services (S3, Lambda, KMS)

Serverless and managed services are inherently highly available — replicate data and functions across multiple AZs by default. Data storage, processing pipeline, and encryption keys are all resilient to AZ failures without additional configuration.

Secure Networking

Redundant VPC Endpoints in each private subnet ensure connectivity to AWS services remains available even if one endpoint fails. All PHI data stays within the AWS network — never traverses the public internet.

Auto-Scaling & Elasticity

- Amazon S3: Scales automatically to accommodate any volume of incoming medical imaging data without pre-provisioning
- AWS Lambda: Scales based on number of images uploaded — from one to thousands of concurrent executions
- SageMaker Auto Scaling: Monitors invocation rate, adds GPU instances during peak hours, removes during quiet periods to reduce costs
- EC2 G5 Auto Scaling Group: Max 50 GPU instances for rapid scaling during diagnostic peaks

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HIPAA Compliance

- 100% encryption at rest using AWS KMS on all S3 buckets and databases
- 100% encryption in transit using TLS for all endpoints
- HIPAA-sensitive data, ML training, and inference in separate isolated accounts
- Mandatory HIPAA-Status tagging on all resources
- OU-level SCPs enforce encryption standards and audit logging
- VPC Endpoints keep PHI data within the AWS network

IaC Implementation

- Terraform provisions all core components — repeatable, non-impacting deployment across dev and prod accounts
- Auto Scaling Group configured for max 50 GPU instances
- S3 enforces 100% KMS encryption at rest and mandatory HIPAA-Status tagging
- S3 Intelligent-Tiering enabled — 60% storage cost reduction

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Multi-Account Structure

- OU: Production — PHI Storage/Access and inference workloads
- OU: ML/R&D — Training workloads with Spot instances
- OU: Compliance — Security/Audit central services
- Separate FinOps account for CUR-based cost visibility, budgeting, and forecasting
- HIPAA-sensitive data isolated in dedicated accounts with strict boundary control

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Key Metrics

- 45% reduction in monthly AI inference costs
- 52% reduction in cost-per-scan (unit economics)
- 35% reduction in ML training costs via SageMaker Spot Training
- R&D provisioning time: 4-6 weeks reduced to less than 1 hour
- Savings Plans coverage for GPU fleet: 10% to 80%
- 100% HIPAA compliant encryption at rest and in transit

ARCHITECTURE 4: KIWIKISAN (AgriTech)

Overview

Customer: KiwiKisan

Industry: Agriculture Technology (Precision Agriculture / IoT)

Engagement: FinOps & IoT Platform Transformation

AWS Account Structure: Multi-account with Prod, Data/IoT Processing, and Shared Services OUs

KiwiKisan operates a precision agriculture platform that ingests data from millions of IoT sensors deployed in agricultural fields. The architecture is 100% serverless and event-driven — eliminating single points of failure and idle compute costs entirely. The design handles high-volume sensor data ingestion, real-time processing, and analytics with sub-5-minute alert latency.

AWS Services Used

AWS Service	Role in Architecture
AWS IoT Core	Secure device connectivity for millions of IoT sensors. Scales automatically to handle connections from millions of devices and process billions of messages without pre-provisioning.
Amazon Kinesis Data Streams	Real-time data streaming from IoT sensors. Shard count set to 50 for high-throughput ingestion. Scales by adjusting shard count to handle any volume of sensor data.
Amazon Kinesis Data Analytics	Real-time stream processing and analytics on sensor data.
AWS Lambda	Event-driven processing — concurrency scales automatically based on data volume in Kinesis stream. Scales from zero to thousands of concurrent executions.
Amazon Timestream	Serverless time-series database for IoT sensor data. Automatically scales ingestion, storage, and query capacity. 55% storage cost reduction vs traditional databases.
Amazon SageMaker	ML models for predictive analytics (crop health, weather patterns). Savings Plans coverage raised from 0% to 75%.
Amazon S3	Long-term storage with lifecycle policies for historical sensor data. Tiered storage for cost optimization.
AWS Budgets & Organizations	Centralized cost governance, mandatory tagging, spend guardrails, anomaly alerts, CUR-based cost visibility.
Aquila Cloud FinOps Platform	Anomaly detection, optimization recommendations, unit cost calculation (cost-per-acre monitored).

High Availability & Resilience

The KiwiKisan architecture is inherently resilient due to its 100% serverless, event-driven design which eliminates single points of failure.

Core Services
The entire data pipeline — AWS IoT Core, Amazon Kinesis Data Streams, AWS Lambda, Amazon Timestream, and Amazon S3 — is built on AWS-managed, regional services.

Multi-AZ by Default
All services are designed for high availability by default, automatically distributing the workload and replicating data across multiple Availability Zones within the AWS Region.

Fault Tolerance
If a single AZ becomes unavailable, these services continue to operate from the healthy AZs without disruption to the data pipeline. There are no EC2 instances to manage or database failovers to configure — the entire workflow is highly resilient with minimal operational overhead.

Auto-Scaling & Elasticity

The solution automatically and cost-effectively scales from a few thousand to millions of IoT sensors without manual intervention.

- AWS IoT Core: Scales automatically to handle secure connections from millions of devices and process billions of messages without pre-provisioning
- Amazon Kinesis Data Streams: Throughput scaled by adjusting shard count (set to 50). Handles any volume of incoming sensor data — from trickles to torrents
- AWS Lambda: Concurrency scales automatically based on data volume in Kinesis stream. Invokes as many parallel instances as needed — from zero to thousands
- Amazon Timestream: Serverless — automatically scales ingestion, storage, and query capacity based on workload. Consistently fast performance even as historical data volume grows

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IaC Implementation

- Terraform defines the entire serverless data pipeline (Kinesis/Lambda) — repeatable, non-impacting deployment
- Kinesis shard_count set to 50 for scaling to millions of IoT sensors
- 100% serverless architecture — eliminates compute waste
- 4 mandatory FinOps tags enforced (Pipeline, CostCenter, Environment, Owner) for accurate unit economics

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Multi-Account Structure

- OU: Prod — Production IoT ingestion and analytics
- OU: Data/IoT Processing — Separate workload accounts for ingestion, processing, and ML
- OU: Shared Services — Central networking, logging, security
- Dedicated FinOps account for consolidated cost reporting, pipeline visibility, and budgeting
- Governance centralized through OU-level controls: mandatory tagging, spend guardrails, anomaly alerts, CUR-based cost visibility

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Key Metrics

- 40% reduction in monthly data processing costs
- 48% reduction in unit cost (cost-per-acre monitored)
- 55% reduction in storage costs via Timestream and S3 lifecycle policies
- Alert latency improved from 2-4 hours to less than 5 minutes
- SageMaker Savings Plans coverage: 0% to 75%
- 100% serverless pipeline — operational overhead reduced from 2 FTEs to 0.5 FTEs

CROSS-ARCHITECTURE COMPARISON

The following table provides a side-by-side comparison of all four architectures for quick auditor reference.

Aspect	Franciscan (EdTech)	Rev (Web3)	Genepowerx (HealthTech)	KiwiKisan (AgriTech)
Primary Compute	EKS + Aurora Serverless	EKS + EC2 Graviton Validators	SageMaker + EC2 G5 GPU	100% Serverless (Lambda + Kinesis)
HA Strategy	Multi-AZ ALB + EKS + RDS failover	Global Accelerator + Multi-AZ + DynamoDB Global Tables	Multi-AZ SageMaker + Redundant VPC Endpoints	Inherent (all serverless, multi-AZ by default)
Scaling	EKS HPA + Cluster Autoscaler + Aurora ACU scaling	EKS dual-layer + DynamoDB on-demand + ASG to 25 nodes	SageMaker Auto Scaling + ASG to 50 GPU instances	IoT Core + Kinesis shards + Lambda concurrency
IaC	Terraform (Aurora Serverless, 4 tags)	Terraform (Graviton, 5 tags, max 25 nodes)	Terraform (GPU ASG, KMS encryption, HIPAA tags)	Terraform (Kinesis shards=50, 4 tags)
Unit Economics	Cost-per-DAU (student)	Cost-per-1M transactions	Cost-per-scan analyzed	Cost-per-acre monitored
Tag Compliance	58% → 96%	45% → 98%	Enforced via SCPs	Enforced via SCPs
Cost Reduction	~30% monthly	34% monthly	45% inference costs	40% processing costs
Compliance	EdTech governance	Web3 financial controls	HIPAA (100% encryption)	IoT data governance
RTO/RPO	4 hrs / 1 hr	4 hrs / 1 hr	4 hrs / 1 hr	4 hrs / 1 hr

COMMON ARCHITECTURAL PATTERNS ACROSS ALL 4 CLIENTS

The auditor should note these consistent patterns across all four architectures:

- Multi-Account Strategy: All 4 clients use AWS Organizations with separate OUs for Production, Non-Production/R&D, and Shared Services — aligned with AWS multi-account best practices
- IaC via Terraform: All infrastructure is defined, deployed, and managed through version-controlled Terraform with CI/CD pipelines
- Multi-AZ Deployment: All architectures deploy across multiple Availability Zones for high availability
- Auto-Scaling: Every architecture uses appropriate auto-scaling mechanisms — EKS HPA/Cluster Autoscaler, Lambda concurrency, DynamoDB on-demand, SageMaker Auto Scaling, Aurora Serverless ACUs
- Mandatory Tagging: All clients have mandatory cost allocation tags enforced via SCPs and Tag Policies
- Unit Economics: Each client has a defined unit economics metric tied to their business (cost-per-DAU, cost-per-transaction, cost-per-scan, cost-per-acre)
- CUR + Aquila Cloud: All clients use AWS CUR integrated with Aquila Cloud FinOps Platform for cost analysis, anomaly detection, and optimization
- Governance: SCPs, Tag Policies, AWS Budgets, and AWS Config are consistently deployed across all engagements

- Security: Encryption at rest and in transit enforced across all architectures. IAM least privilege. CloudTrail enabled in all regions with immutable S3 logging
- RTO/RPO: Consistent 4-hour RTO and 1-hour RPO targets across all clients